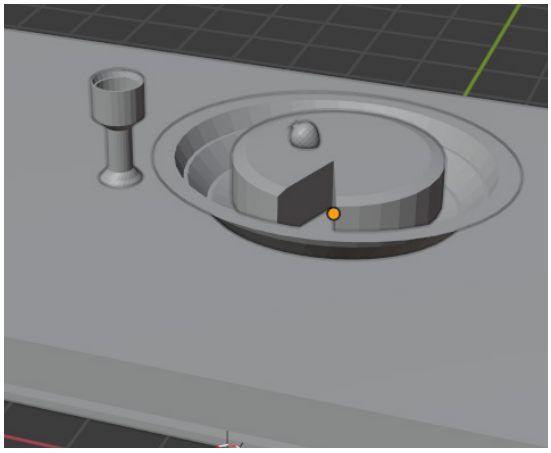
Blender Modeling Modifiers



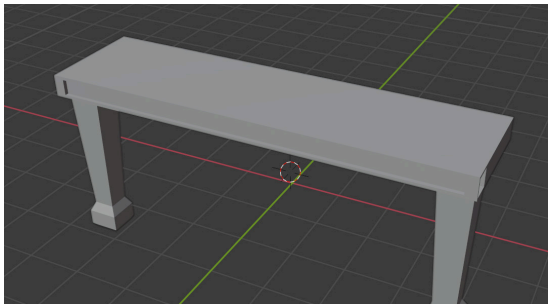
At this point we can do some basic polygon modeling in Blender using Edit Mode and moving, scaling and rotating verities, edges and faces. In this lab we will continue our modeling practice by learning how to use Blender modifiers. Blender modifier are additional tools we can use to modify and create geometry in Blender. Think of modifiers in Blender like layers styles in Adobe Photoshop or Effects in Adobe Illustrator.

In this lab we are going to model a lovely dinning table with dessert on it! We'll learn how to use the mirror, subdivision surface, boolean and screw modifiers to help us build a scene. In this scene we are modeling a table that has a piece of cake and plate on it.

Setup

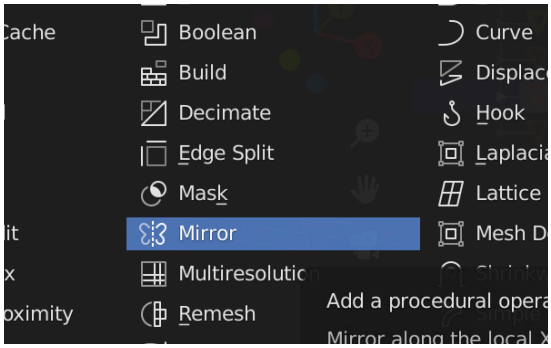
Begin by opening *modifiers.blend* into Blender. You'll notice a half table and two thin extrude paths.

Mirror Modifier



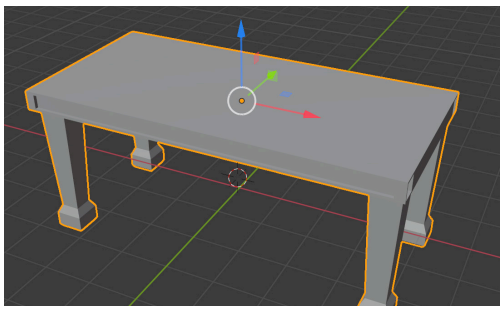
You'll notice that the table is somewhat intricate, and only modeled as halfway. Many 3D meshes are symmetrical and Blender makes it easy to mirror the a 3D mesh to save time.

Select the table mesh and go to the properties panel. Click the modifier menu icon which looks like a wrench. By default you will not have any modifiers. Click the button that says '*Add Modifier*'. From the dialogue box under Generate choose '*Mirror*'.



The mirror modifier will appear in the properties panel. If you wish to remove a modifier click the ***x*** on the far right corner. To turn off visually a modifier click the computer icon called *realtime*. This does not delete the modifier, but will turn it off until you enable it again.

With the table selected choose the y-axis, or mirror the object along the y-axis of the object, completing the shape. Click the realtime button again to turn on and off the second half of the table.

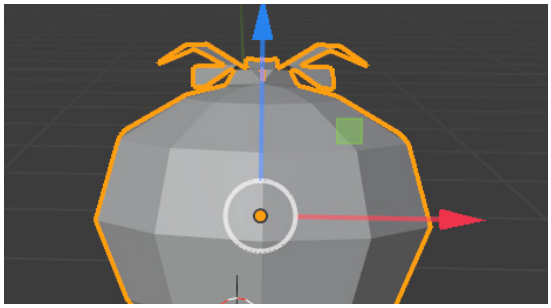


Put the table in *Edit Mode* and adjust the geometry of the table, you'll see that the mirror side reflects any changes that you make.

To be able to adjust the geometry in the mirrored section on it's own click the table and hit the '*Apply*' button. This will apply the mirror to your mesh giving you both sides of the table. Switch to

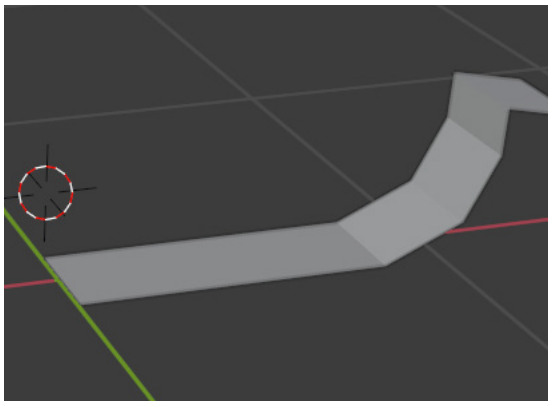
edit mode and you'll see that you can adjust geometry on one side of the table without affecting the other side of the table.

Click on the strawberry in the scene and turn on the visibility. Mirror it along the x-axis, and hit *apply*. Moe the strawberry onto the table.



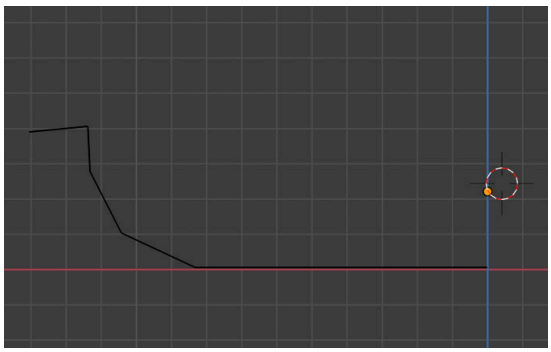
Screw Modifier

Some objects we want to model are rounded or cylindrical-type objects. These objects in the real worlds might be made using a lathe machine or similar type machine for glass. Think of it like a slice of cake. It you have one slice, and then add all the slices of the cake together you get a circle or cylinder.

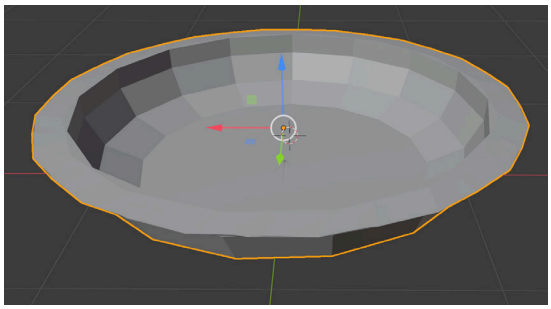


In Blender we can use a modifier called the ***screw*** modifier. It allows us to take a single strip of geometry and spin it around an axis to form a 360˚ mesh.

Begin by selecting *Plate-Slice* in the outliner and turning it on. Select the *Plate-Slice* and go into *Edit Mode*. Then put then scene in Front Orthographic View. You'll see a black line that looks like the image on the right, this is your *Plate-Slice*. This was created by taking the edge of plane and extruding it to form this simple wireframe. What we will do is take this wireframe and multiple it and revolve it around the z-axis.



Go back to *Object Mode* and select the *Plate-Slice*. Go to the modifier menu and choose the screw modifier. Make sure that the model is rotating around the z-axis. Look at your scene and it should look like a plate or saucer. You can go in object mode if you like to scale the object to stretch or squash the mesh.

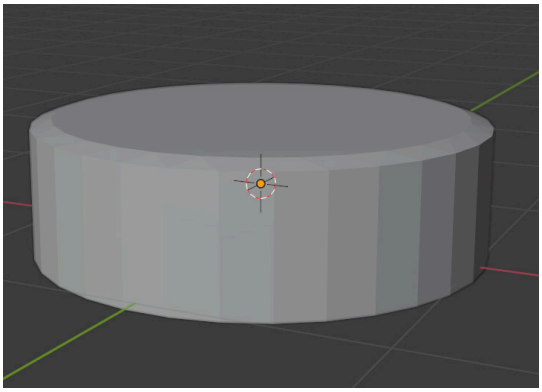


Finally hit the '*Apply'* button, this will give the plate-slice geometry that can be manipulated in *Edit Mode*.

Click on *Cup-Slice* in the Outliner. Make it visible and use the screw modifier to revolve the geometry slice around the z-axis.

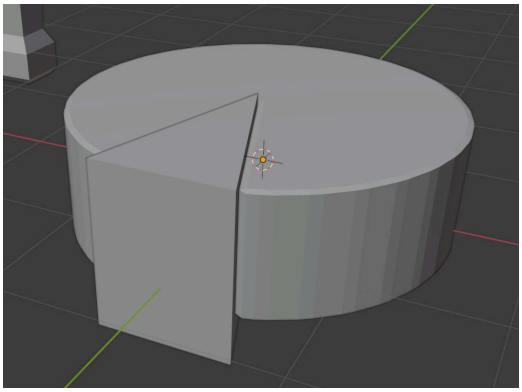
Add the plate and cup on top of the table.

Bevel / Boolean Modifier



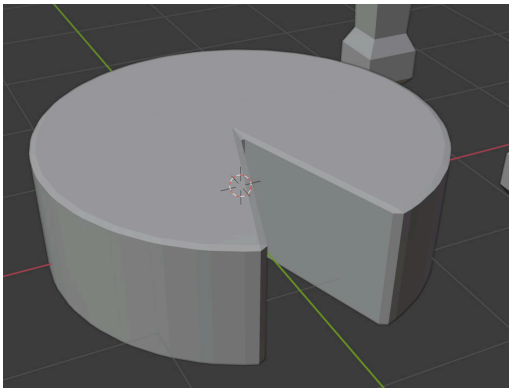
Next we are going to model a cake for our scene. We'll begin by adding a cylinder by adding *Mesh > Add > Cylinder*. In your outliner label the cylinder '*cake*'. Use the scale tool to make the cylinder a bit more squat and cake-like. Let's make this look a bit nicer by adding a *bevel* modifier. The bevel modifier is a bit like the bevel tool, the difference is the modifier can always be selected and adjusted.

Select the cake mesh and add the bevel modifier from the menu. You'll see it applies a simple bevel that makes your mesh a bit more cake-like. I used an offset of about 0.15m. Use the *apply* button to apply the modifier to the cake permanently.



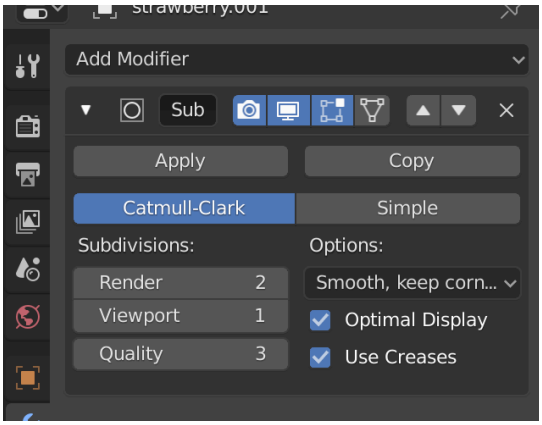
Now let's cut a slice out of the cake. In the outliner turn on the mesh called *triangle*. Use the extrude tool to make the taller. Overlap the extruded triangle with the cake as shown in the image on the right.

What we will do now is use a *boolean* modifier to cut the extruded triangle from the cake. The boolean modifier is a bit like the pathfinder tool in Adobe Illustrator. The pathfinder tool in Adobe Illustrator allows you to add or subtract vector paths together, Blender is similar but allows you to add, subtract or intersect meshes with one another. In this case we will subtract the extrude triangle from the cake.



Begin by selecting the cake object mesh and add the boolean modifier. Use the object search to locate the triangle, or use the eyedropper tool in Blender and select the extrude triangle. Set the operation to difference, which will subtract the extruded triangle from the cake. Hit apply and you should see a slice taken out of your cake.

Subdivision Surface

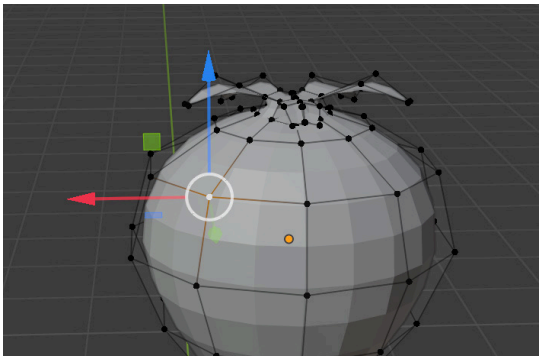


So far our models have been fairly angular. That is fine for more angular type objects like houses and tables, but what about more organic objects like animals or a bowl?

You could add more geometry like edges and verticies with loop cuts to more around geometry, but that can be time consuming and difficult. Blender has an easier way of creating more rounded objects using a modifier called subdivision surfaces.

Subdivisional surfaces round out objects giving them more geometry and more curves. Before we use a subdivision surface modifier please BE CAREFUL. Adding too much geometry may crash Blender or even your computer, so save your work before you use the subdivision surface modifier.

Select the strawberry in the scene and add the subdivision surface modifier, you see that is taking a relatively angular model and rounding it out to smooth.



If you go into Edit Mode you'll notice a cage surrounding the strawberry. By editing the cage you'll will push, pull and model the strawberry shape. To fully apply the subdivision surface switch back into object mode and click the 'Apply' button in the modifier to apply the subdivision surface permenantely.

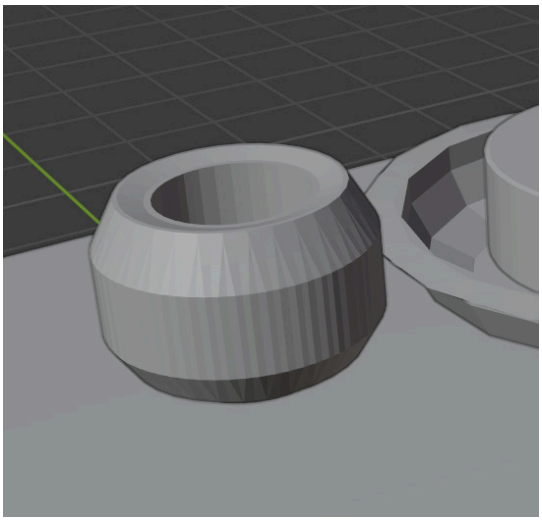
Apply a subdivision surface modifier to the cup and plate.

Position the plate on the table. Then position the cake on top of the plate and the strawberry on top of the cake. Place the cup to the side of the cake. Viola! One model dessert spread!

Save your file as *lastname-firstname-modifier.blender*

Extra Credit +2 points (not-required)

Create a vase by creating a geometric slice and using the screw modifier. After creating the vase with the screw modifier use the subdivision surface modifier to round it out.



Place the vase on the table next to the cake.